Command Execution via Rmd Syntax Highlighting in Vim < 9.2.1178

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Content

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- Advisory: GHSA-x2w5-pj55-7hpc
- Severity: medium
- Published: 2026-10-12
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Affected

- `vim` (vim): < v9.2.1178

Description

Command Execution via Rmd Syntax Highlighting in Vim < 9.2.1178 ========== Date: 12.10.2026 Severity: Medium CVE: *requested, not yet assigned* CWE: Improper Neutralization of Special Elements used in an OS Command (CWE-78), Improper Input Validation (CWE-20)

Summary

The R Markdown syntax script in `runtime/syntax/rmd.vim` collects the language names of code chunks from the buffer and passes them to `globpath()`. The script strips non-word characters from each name with a regular expression, which a crafted byte sequence can make stop early, so text from the file reaches `globpath()` unfiltered. On Unix, `globpath()` hands such a pattern to the shell, which allows arbitrary command execution when the file is opened.

Description

With the default `g:rmd_dynamic_fenced_languages`, `s:CheckRmdFencedLanguages()` reads the language name of every code chunk in the buffer and calls `s:IncludeLanguage()` for names it has not seen yet, which checks for a matching syntax file using `globpath()`.

The name was trimmed with `substitute(v:val, '\W.*', '', '')`. The NFA regexp engine treats an overlong UTF-8 encoding of NUL as the end of the text, so the pattern does not match and the rest of the name is kept. If the name contains backticks, `globpath()` passes it to the shell for expansion.

Impact

Arbitrary command execution in the context of the user running Vim, when the user opens a crafted R Markdown or Quarto file (`.rmd`, `.smd`, `.qmd` and similar).

Exploitation requires:

- a Unix-like system (on Windows `globpath()` does not invoke the shell),
- syntax highlighting enabled
- the victim opening a crafted file.

The severity is rated Medium because the user must open a crafted file of a specific rmd file type.

Acknowledgements

The Vim project would like to thank github user tylerni7-ant for reporting and analyzing the issue.

References

The issue has been fixed as of Vim patch v9.2.1178.

- Commit
- Github Security Advisory

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